

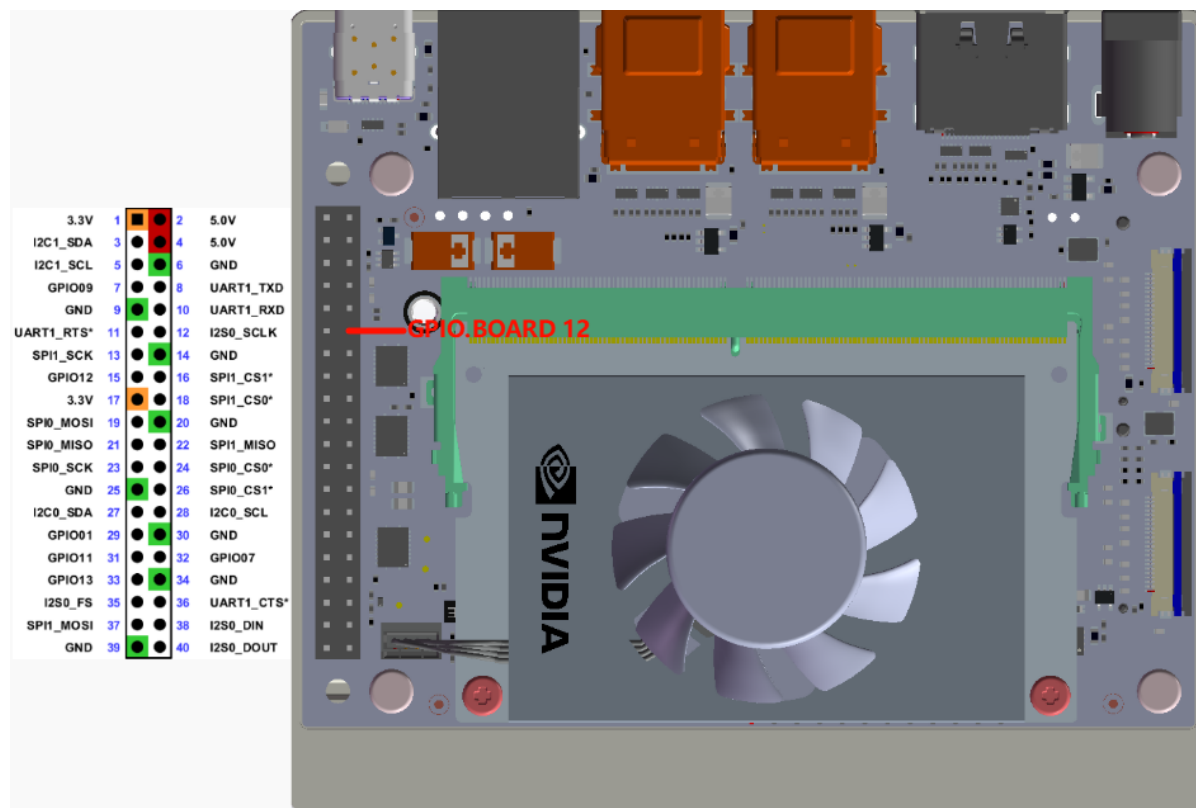
GPIO output function

GPIO output function

1. GPIO pin diagram comparison table
2. Jetson orin super GPIO repair
 1. Busybox modifies registers (taking physical pin 31 as an example)
 2. Change the device tree (taking the change of physical pin 7 as an example)
3. Program effect

1. GPIO pin diagram comparison table

GPIO.BOARD 12 pin corresponds to GPIO.BCM 18 pin:



2. Jetson orin super GPIO repair

Note: After upgrading to Jetson orin super, most GPIO ports have changed from the original bidirectional (input + output) to input only. Now two solutions are provided to modify GPIO to a bidirectional interface.

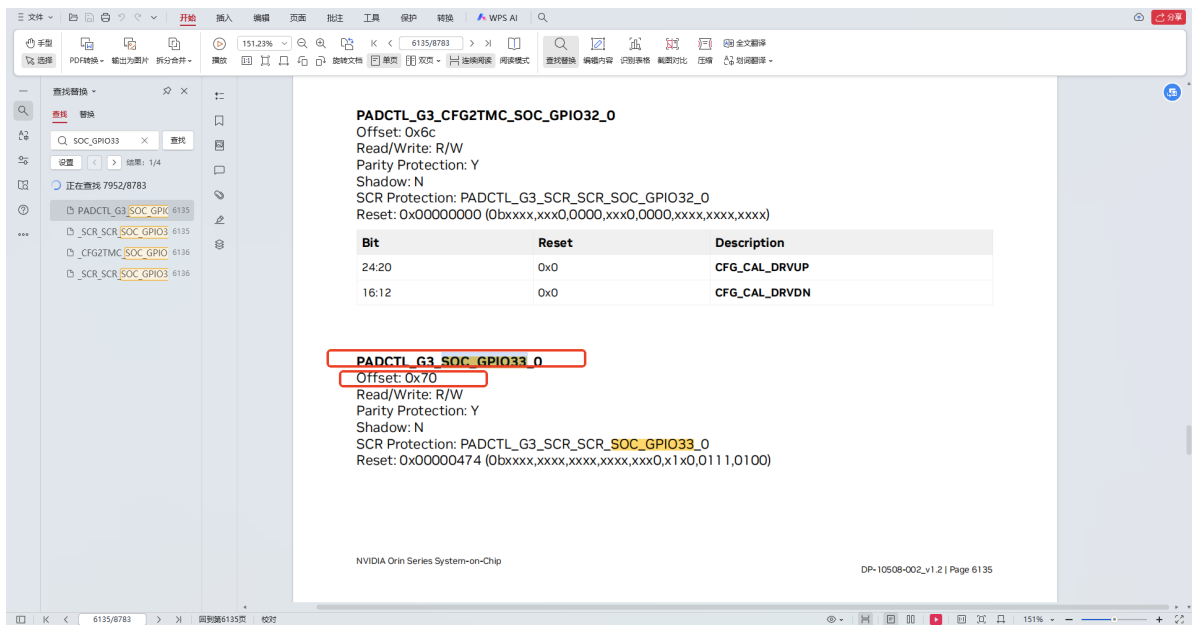
1. Busybox modifies registers (taking physical pin 31 as an example)

This solution is a temporary solution. After modifying the register configuration, it will become invalid every time it is restarted. If you need to keep it in effect, refer to the second solution.

The documents in the picture can be found in the attachment - jetson gpio, which are all found and downloaded from the official website (software data\data documents\jetson-gpio).

After inquiry, it is found that: physical pin 31 ---- corresponds to GPIO11--- corresponds to SOC_GPIO33, the corresponding **PADCTL_G3** address = **PADCTL_A0** is 0x2430000, and the corresponding SOC_GPIO33 is 0x2430070 (refer to the following picture)

Sysfs GPIO	Name	Pin	Pin	Name	Sysfs GPIO
Jetson Xavier NX J12 Header					
	3.3 VDC Power	1	2	5.0 VDC Power	
	I2C1_SDA I2C Bus 8	3	4	5.0 VDC Power	
	I2C1_SCL I2C Bus 8	5	6	GND	
148 gpio436	GPIO9 AUDIO_MCLK	7	8	UART1_TX /dev/ttyTHS0	
	GND	9	10	UART1_RX /dev/ttyTHS0	
140 gpio428	UART1_RTS	11	12	I2S0_SCLK	157 gpio445
192 gpio480	SPI1_SCK	13	14	GND	
20 gpio268	GPIO12 Alt: PWM	15	16	SPI1_CS1	196 gpio484
	3.3 VDC Power	17	18	SPI1_CS0	195 gpio483
205 gpio493	SPI0_MOSI	19	20	GND	
204 gpio492	SPI0_MISO	21	22	SPI1_MISO	193 gpio481
203 gpio491	SPI0_SCK	23	24	SPI0_CS0	206 gpio494
	GND	25	26	SPI0_CS1	207 gpio495
	I2C0_SDA I2C Bus 1	27	28	I2C0_SCL I2C Bus 1	
133 gpio421	GPIO01	29	30	GND	
134 gpio422	GPIO11	31	32	GPIO07 Alt: PWM	136 gpio424
105 gpio393	GPIO13 Alt: PWM	33	34	GND	
160 gpio448	I2S0_FS	35	36	UART1_CTS	141 gpio429
194 gpio482	SPI1_MOSI	37	38	I2S0_SDIN	159 gpio447
	GND	39	40	I2S0_SDOUT	158 gpio446



Pinmux register address:

System Components

- Multi-Purpose I/O Pins and Pin Multiplexing (PinMux) Registers
 - G3 PAD Control Registers; Updated note to use Block Name (PADCTL_A0) to find the Base Address of these registers in the System Address Map
 - UFS PAD Control Registers; Updated note to use Block Name (PADCTL_A17) to find the Base Address of these registers in the System Address Map
- Design for Debugging (DFD); Updates to FSI TSC section: added FSI_CSITE_TSINTP Render Configuration, Connectivity, Calculating Time-Interval Using Timestamps subsections
- Always-On Cluster (AON) and SPE; new chapter

PADCTL_A	0x02430000	0x0244ffff	SYSTEM
PADCTL_A0	0x02430000	0x02430fff	SYSTEM
PADCTL_A2	0x02432000	0x02432fff	SYSTEM
PADCTL_A4	0x02434000	0x02434fff	SYSTEM
PADCTL_A5	0x02435000	0x02435fff	SYSTEM
PADCTL_A6	0x02436000	0x02436fff	SYSTEM
PADCTL_A7	0x02437000	0x02437fff	SYSTEM
PADCTL_A8	0x02438000	0x02438fff	SYSTEM
PADCTL_A11	0x0243b000	0x0243bfff	SYSTEM
PADCTL_A13	0x0243d000	0x0243dfff	SYSTEM
PADCTL_A16	0x02440000	0x02440fff	SYSTEM
PADCTL_A17	0x02441000	0x02441fff	SYSTEM
PADCTL_A20	0x02444000	0x02444fff	SYSTEM
PADCTL_A21	0x02445000	0x02445fff	SYSTEM
PADCTL_A24	0x02448000	0x02448fff	SYSTEM
PADCTL_A25	0x02449000	0x02449fff	SYSTEM

- The Offset is 0x70.
- The Pinmux register address is 0x2430070

1.1 Install busybox

```
sudo apt install busybox
```

1.2 Set the gpio pin to output:

```
sudo busybox devmem 0x02430070 w 0x004
```

1.3 Test output

GPIO and BCM comparison table

BCM code	Function name	Physical pin		Function name	BCM code
	3V3	1	2	5V	
2	SDA	3	4	5V	
3	SCL	5	6	GND	
4	D4	7	8	D14(TXD)	14
	GND	9	10	D15(RXD)	15
17	D17	11	12	D18	18
27	D27	13	14	GND	
22	D22	15	16	D23	23
	3V3	17	18	D24	24
10	D10	19	20	GND	
9	D9	21	22	D25	25
11	D11	23	24	D8	8
	GND	25	26	D7	7
0	DO(ID_SD)	27	28	D1(ID_SC)	1
5	D5	29	30	GND	
6	D6	31	32	D12	12
13	D13	33	34	GND	
19	D19	35	36	D16	16
26	D26	37	38	D20	20
	GND	39	40	D21	21

```
cd ~/jetson-gpio/samples

export JETSON_MODEL_NAME=JETSON_ORIN_NX

python3 simple_out.py
```

After the setting is completed, refer to the BCM code, change the code in the test case to the corresponding value, and then run the following test case to find the physical pin 31 Switching high and low levels back and forth

2. Change the device tree (taking the change of physical pin 7 as an example)

Note: This solution requires a certain understanding of the device tree and is not recommended for novices. After this solution is successful, the set gpio pin will become a bidirectional pin, that is, input + output.

Physical pin 7 corresponds to GPIO 09, corresponds to SOC_GPIO59, and the corresponding device tree name is soc_gpio59_pac6

4	A	B	C	D	V	W	X	Y	Z	AA	AB	AC	AD	
1														
2														
3	Revision 1.04													
4														
5														
6														
7	SODIMM Pin #	Jetson SODIMM Signal Name	Jetson Orin NX and Nano Function	Verilog Ball Name	Device Tree Pin Name	Power Rail	Waku	Strap	CTE	DP Central	DP Group	Pad Category	Pad Type	S
107	100	DP1_AIN_F	DP1_AIN_F	DP_AIN_C00_F	dp_ain_c00_f	+ddc_vddq1						DP_AIN	DIFFPAIR_LPC_V0101010K	N/A
108	96	DP1_AIN_B	DP1_AIN_B	DP_AIN_C00_B	dp_ain_c00_b	+ddc_vddq1						DP_AIN	DIFFPAIR_LPC_V0101010K	N/A
109														
110	96	DP1_AIN_B	DP1_AIN_B	DP_AIN_C00_B	dp_ain_c00_b	+ddc_vddq1						DP_AIN	DIFFPAIR_LPC_V0101010K	N/A
124	218	QTD12	QTD12	SOC_QTD09	soc_gpio59_pac6	+ddc_vddq				QF_PWM1	QF_PWM1	QF_PWM1	DIFFPAIR_LPC_V0101010K	N/A
126														
127	214	FORCE_RECOVER*	FORCE_RECOVER*	SOC_QTD013	soc_gpio13_p00	+ddc_vddq		rom_0		Q4	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
133	206	QTD07	QTD07 (PMM)	SOC_QTD019	soc_gpio19_p00	+ddc_vddq				QF_PWM7	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
135	228	QTD13	QTD13 (PMM)	SOC_QTD021	soc_gpio21_p00	+ddc_vddq				QF_PWM5	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
141	114	CARD_PVDR	CARD_PVDR	UART4_CTS	uart4_cts_p00	+ddc_vddq				Q4	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
142	199	ICSD0_SCLX	ICSD0_SCLX	SOC_QTD041	soc_gpio41_p07	+ddc_vddq				Q4	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
143	193	ICSD0_J00T	ICSD0_J00T	SOC_QTD042	soc_gpio42_p10	+ddc_vddq				Q4	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
144	196	ICSD0_J00K	ICSD0_J00K	SOC_QTD043	soc_gpio43_p11	+ddc_vddq				Q4	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
145	197	ICSD0_J00L	ICSD0_J00L	SOC_QTD044	soc_gpio44_p12	+ddc_vddq				Q4	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
146	232	ICSD0_SCL	ICSD0_SCL	GPIO1_ICSD0_SCL	gpio1_icd0_scl_p13	+ddc_vddq				Q4	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
147	234	ICSD0_SDA	ICSD0_SDA	GPIO1_ICSD0_SDA	gpio1_icd0_sda_p14	+ddc_vddq				Q4	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
188														
189	120	CARD_PVDR	CARD_PVDR	GPIO1_ICSD0_SDA	gpio1_icd0_sda_p14	+ddc_vddq				QF_PWM1	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
195	211	QTD109	QTD109 (AUX_NCTLA)	SOC_QTD009	soc_gpio09_pac6	+ddc_vddq				QF_PWM1	Q4	Q4	DIFFPAIR_LPC_V0101010K	N/A
204														
207	182	PCIE1_CLKREQ*	PCIE1_CLKREQ*	PEX_L1_CLKREQ_B	per_l1_clkreq_b_p02	+ddc_vddc_ctl				PEX_CTL	PEX_CTL_A	Q4	DIFFPAIR_LPC_V0101010K	N/A
208	183	PCIE1_RST*	PCIE1_RST*	PEX_L1_RST_B	per_l1_rst_b_p03	+ddc_vddc_ctl				PEX_CTL	PEX_CTL_A	Q4	DIFFPAIR_LPC_V0101010K	N/A
213	180	PCIE0_CLKREQ*	PCIE0_CLKREQ*	PEX_L0_CLKREQ_B	per_l0_clkreq_b_p10	+ddc_vddc_ctl				PEX_CTL	PEX_CTL_A	Q4	DIFFPAIR_LPC_V0101010K	N/A
214	181	PCIE0_RST*	PCIE0_RST*	PEX_L0_RST_B	per_l0_rst_b_p11	+ddc_vddc_ctl				PEX_CTL	PEX_CTL_A	Q4	DIFFPAIR_LPC_V0101010K	N/A
215	179	PCIE0_WAKEN	PCIE0_WAKEN	PEX_WAKEN_B	per_wake_b_p12	+ddc_vddc_ctl				PEX_CTL	PEX_CTL_A	Q4	DIFFPAIR_LPC_V0101010K	N/A

Put the jetson-orin-gpio-patch-main file in software data\data document\jetson-gpio into jetson orin, open the pin7_as_gpio.dts file, and change the circled area to the GPIO pin you need to change

```

dts-v1;
/plugin/;

/ {
    jetson-header-name = "Jetson 40pin Header";
    overlay-name = "pin7_as_gpio bidirectional";
    compatible = "nvidia,p3768-0000+p3767-0000\nvidia,p3768-0000+p3767-0003\nvidia,p3768-0000+p3767-0004\nvidia,p3768-0000+p3767-0005\nvidia,p3768-0000+p3767-0006\nvidia,p3768-0000+p3767-0007\nvidia,p3768-0000+p3767-0008\nvidia,p3768-0000+p3767-0009\nvidia,p3768-0000+p3767-0010\nvidia,p3768-0000+p3767-0011\nvidia,p3768-0000+p3767-0012\nvidia,p3768-0000+p3767-0013\nvidia,p3768-0000+p3767-0014\nvidia,p3768-0000+p3767-0015\nvidia,p3768-0000+p3767-0016\nvidia,p3768-0000+p3767-0017\nvidia,p3768-0000+p3767-0018\nvidia,p3768-0000+p3767-0019\nvidia,p3768-0000+p3767-0020\nvidia,p3768-0000+p3767-0021\nvidia,p3768-0000+p3767-0022\nvidia,p3768-0000+p3767-0023\nvidia,p3768-0000+p3767-0024\nvidia,p3768-0000+p3767-0025\nvidia,p3768-0000+p3767-0026\nvidia,p3768-0000+p3767-0027\nvidia,p3768-0000+p3767-0028\nvidia,p3768-0000+p3767-0029\nvidia,p3768-0000+p3767-0030\nvidia,p3768-0000+p3767-0031\nvidia,p3768-0000+p3767-0032\nvidia,p3768-0000+p3767-0033\nvidia,p3768-0000+p3767-0034\nvidia,p3768-0000+p3767-0035\nvidia,p3768-0000+p3767-0036\nvidia,p3768-0000+p3767-0037\nvidia,p3768-0000+p3767-0038\nvidia,p3768-0000+p3767-0039\nvidia,p3768-0000+p3767-0040\nvidia,p3768-0000+p3767-0041\nvidia,p3768-0000+p3767-0042\nvidia,p3768-0000+p3767-0043\nvidia,p3768-0000+p3767-0044\nvidia,p3768-0000+p3767-0045\nvidia,p3768-0000+p3767-0046\nvidia,p3768-0000+p3767-0047\nvidia,p3768-0000+p3767-0048\nvidia,p3768-0000+p3767-0049\nvidia,p3768-0000+p3767-0050\nvidia,p3768-0000+p3767-0051\nvidia,p3768-0000+p3767-0052\nvidia,p3768-0000+p3767-0053\nvidia,p3768-0000+p3767-0054\nvidia,p3768-0000+p3767-0055\n";

    fragment@0 {
        target = <gpio-mux>;
        __overlay__ {
            pinctrl-names = "default";
            pinctrl-0 = <jetson_io_pinmux>;
            jetson_io_pinmux: exp-header-pinmux {
                hd40 {
                    nvidia,pins = "soc_gpio59_pac6";
                    nvidia,tristate = "on";
                    nvidia,enable-input = <0x1>;
                    nvidia,pull = <0x0>;
                };
            };
        };
    };
};

```

In jetson-orin-gpio-patch Run the following commands respectively

```

dtc -O dtb -o pin7_as_gpio.dtbo pin7_as_gpio.dts #You can change the generated
file name according to the different GPIO port names
sudo cp pin7_as_gpio.dtbo /boot
sudo /opt/nvidia/jetson-io/jetson-io.py

```

===== Jetson Expansion Header Tool =====

Select one of the following:

Configure Jetson 40pin Header

Configure Jetson 24pin CSI Connector

Configure Jetson M.2 Key E Slot

Exit

===== Jetson Expansion Header Tool =====

3.3V	(1)	..	(2)	5V
i2c8	(3)	..	(4)	5V
i2c8	(5)	..	(6)	GND
unused	(7)	..	(8)	uarta
GND	(9)	..	(10)	uarta
unused	(11)	..	(12)	unused
unused	(13)	..	(14)	GND
unused	(15)	..	(16)	unused
3.3V	(17)	..	(18)	unused
unused	(19)	..	(20)	GND
unused	(21)	..	(22)	unused
unused	(23)	..	(24)	unused
GND	(25)	..	(26)	unused
i2c2	(27)	..	(28)	i2c2
unused	(29)	..	(30)	GND
unused	(31)	..	(32)	unused
unused	(33)	..	(34)	GND
unused	(35)	..	(36)	unused
unused	(37)	..	(38)	unused
GND	(39)	..	(40)	unused

Jetson 40pin Header:

Configure for compatible hardware
Configure header pins manually
Back

===== Jetson Expansion Header Tool =====

Select one of the following options:

Adafruit SPH0645LM4H

Adafruit UDA1334A

~~FE PI Audio V1 and Z V2~~

Pin 29 gpio bidirectional

ReSpeaker 4 Mic Array

ReSpeaker 4 Mic Linear Array

Back

===== Jetson Expansion Header Tool =====

3.3V	(1)	..	(2)	5V
i2c8	(3)	..	(4)	5V
i2c8	(5)	..	(6)	GND
unused	(7)	..	(8)	uarta
GND	(9)	..	(10)	uarta
unused	(11)	..	(12)	unused
unused	(13)	..	(14)	GND
unused	(15)	..	(16)	unused
3.3V	(17)	..	(18)	unused
unused	(19)	..	(20)	GND
unused	(21)	..	(22)	unused
unused	(23)	..	(24)	unused
GND	(25)	..	(26)	unused
i2c2	(27)	..	(28)	i2c2
unused	(29)	..	(30)	GND
unused	(31)	..	(32)	unused
unused	(33)	..	(34)	GND
unused	(35)	..	(36)	unused
unused	(37)	..	(38)	unused
GND	(39)	..	(40)	unused

Jetson 40pin Header:

=====

```
===== Jetson Expansion Header Tool =====

Select one of the following:
Re-configure Jetson 40pin Header
Configure Jetson 24pin CSI Connector
Configure Jetson M.2 Key E Slot
Save and reboot to reconfigure pins
Save and exit without rebooting
Discard all pin changes
Exit
```

Start the test:

```
cd ~/jetson-gpio/samples
```

```
export JETSON_MODEL_NAME=JETSON_ORIN_NX
```

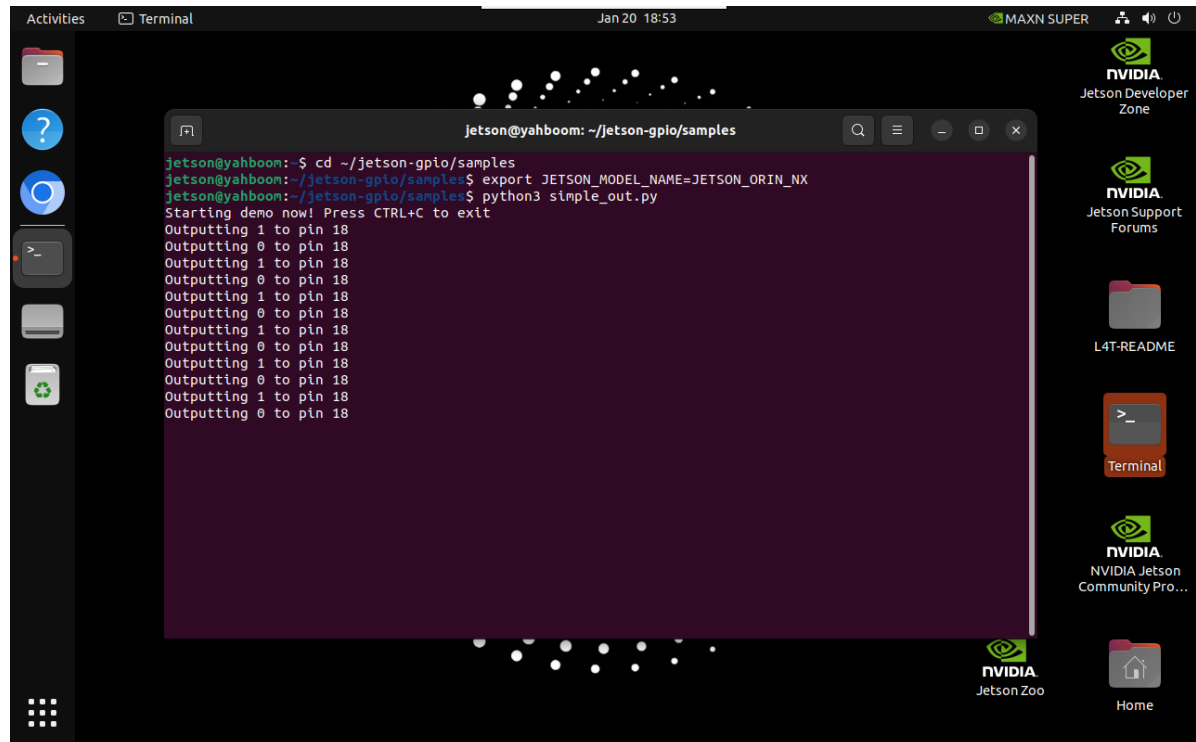
```
python3 simple_out.py
```

After the setting is completed, refer to the BCM code, change the code in the test case to the corresponding value, and then run the following test case, you can find that the physical pin 7 switches back and forth between high and low levels

3. Program effect

Use DuPont wire to lead out GPIO.BOARD 6 and GND pins, and use a multimeter to measure the pin voltage change:

Note: Do not connect incorrectly or cause pin short circuit, otherwise it may cause damage to the motherboard hardware!



The screenshot shows a terminal window titled "jetson@yahboom: ~/jetson-gpio/samples". The terminal output is as follows:

```
jetson@yahboom:~$ cd ~/jetson-gpio/samples
jetson@yahboom:~/jetson-gpio/samples$ export JETSON_MODEL_NAME=JETSON_ORIN_NX
jetson@yahboom:~/jetson-gpio/samples$ python3 simple_out.py
Starting demo now! Press CTRL+C to exit
Outputting 1 to pin 18
Outputting 0 to pin 18
Outputting 1 to pin 18
Outputting 0 to pin 18
Outputting 1 to pin 18
Outputting 0 to pin 18
Outputting 1 to pin 18
Outputting 0 to pin 18
Outputting 1 to pin 18
Outputting 0 to pin 18
Outputting 1 to pin 18
Outputting 0 to pin 18
Outputting 1 to pin 18
Outputting 0 to pin 18
```

The desktop background is dark with a pattern of white dots. On the left is a vertical dock with icons for Activities, a question mark, a blue circle, a terminal, a folder, and a recycling bin. On the right is a sidebar with icons for NVIDIA Jetson Developer Zone, NVIDIA Jetson Support Forums, L4T-README, a terminal icon labeled "Terminal", NVIDIA Jetson Community Pro..., and a Home icon. The top status bar shows "Activities", "Terminal", "Jan 20 18:53", and "MAXN SUPER".